

Mirage

X17 (2017) — English translation

On a hot summer day, when the air warmed up to $t_1 = 35\text{ }^\circ\text{C}$, the temperature of the asphalt in the sun and the thin layer of air above it was $t_2 = 65\text{ }^\circ\text{C}$. When walking along the highway in such dry weather, “puddles” are often visible on the road. The refractive index of air depends linearly on air density: $n = 1 + k\rho$. At $t_0 = 0\text{ }^\circ\text{C}$, $n_0 = 1.000292$. Consider that a person's height is $H = 1.8\text{ m}$.

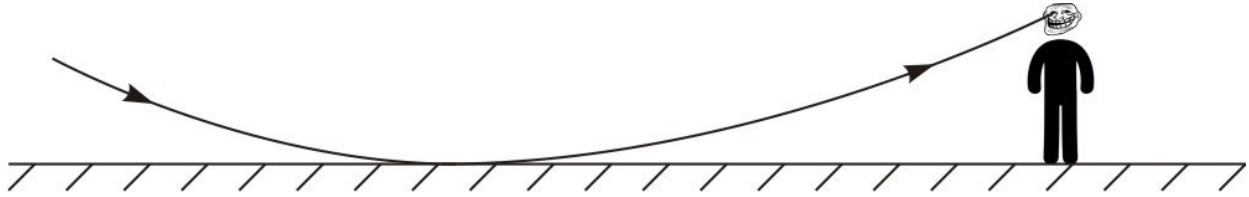


Fig. 1.

A1 At what minimum distance L_{min} can a “puddle” be seen? Express the answer in terms of H , n_0 , T_0 , T_1 , T_2 and calculate the resulting value. **5.00**

Source: <https://pho.rs/p/3942>